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Summary of Work – SDACS, Capstone I

Hardware and Physical Design

- **Component Specification:** The system moved from a generic ESP32 concept to the specific use of the Adafruit Metro S3 (ESP32-S3) for recording nodes, selected for its integrated battery management and I2S support.
- **Sensor Integration:** Specific high-performance components were selected and integrated, including the ICS-43434 digital MEMS microphone for audio capture and the Sensirion SHT41 for environmental sensing.
- **Custom PCB Development:** Comprehensive schematics and a PCB layout were designed for manufacturing, moving beyond the functional block diagrams in the proposal to a production-ready hardware architecture.
- **Enclosure Maturity:** The physical design reached Revision C, featuring a modular assembly consisting of a dedicated battery port, MCU housing, and a secure lid.

Software and Backend Architecture

- **Server Implementation:** The server role was fully defined using Node-RED on a Raspberry Pi 5 to handle MQTT telemetry, data normalization, and database management.
- **Data Management:** A structured data pipeline was established to convert incoming JSON telemetry into SQLite and CSV formats for long-term logging and analysis.
- **System Reliability:** Advanced firmware features were added, including a "heartbeat" monitoring system to detect stale (offline) nodes and the implementation of an Over-the-Air (OTA) firmware update protocol via GitHub.
- **Signal Processing:** On-node processing capabilities were refined to include real-time RMS, dBFS, and Sound Pressure Level (SPL) estimation alongside FFT calculations.

Testing and Validation Results

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- **Acoustic Calibration:** A formal procedure was executed to establish a calibration constant ($K \approx 96.99$ dB), allowing for accurate conversion of digital audio levels into real-world SPL measurements.
- **Functional Success:** System-level testing confirmed successful MQTT communication, audio processing, and SD card data logging.
- **Performance Identification:** Testing revealed that while acoustic measurements were within the ± 3.2 dB error tolerance, temperature sensing was inconsistent, leading to its classification as a "Partial" pass for future refinement.

Project Management and Requirements

- **Refined Requirements:** The broad goals of the proposal were codified into a formal MoSCoW (Must, Should, Could, Won't) table to strictly define project scope and deliverables.
- **Risk Mitigation:** The initial risk list was expanded into a detailed Risk Assessment Matrix, incorporating specific technical challenges such as node synchronization and custom PCB complexity.